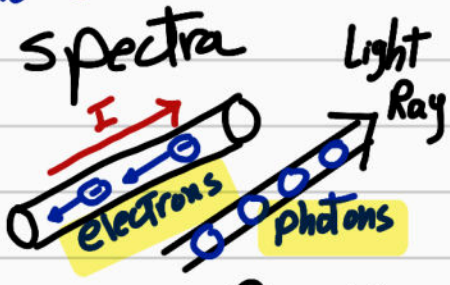
1 Black Body Radiation "Chapter ⑤"

1 Classical Vs Modern:

Classical physics: Branch of physics study daily obervations & common Exp. example: Electricity

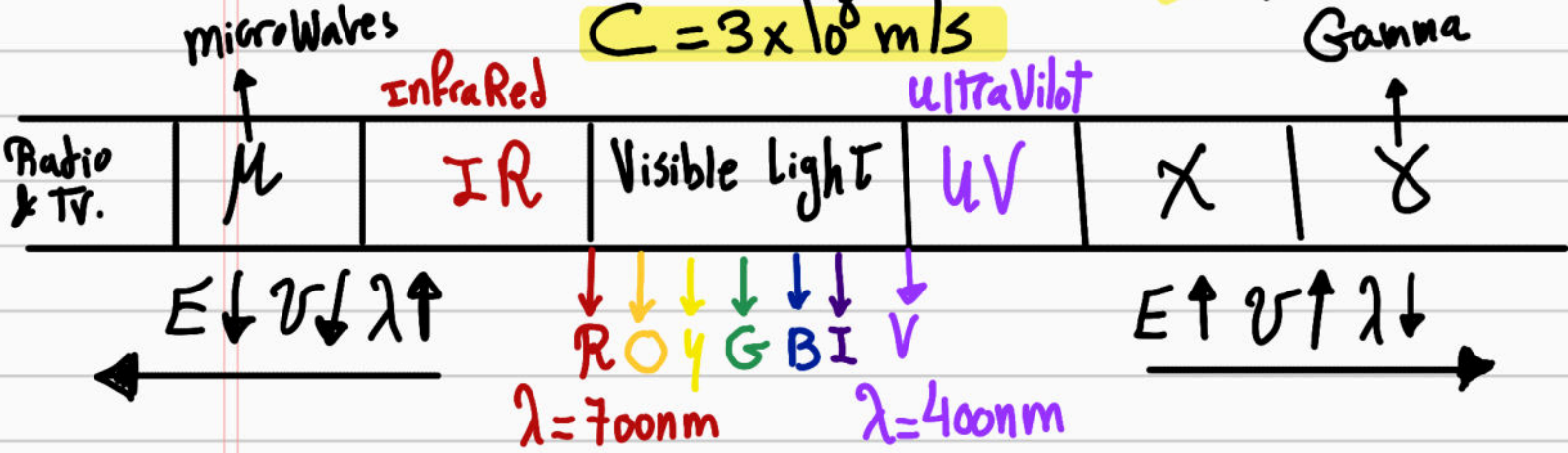
Modern physics: Branch of physics sTudy atomic & sub atomic scale phenomena

example: ATomic spectra



2 ElectroMagnetic spectrum:

$C = 3 \times 10^8$ m/s

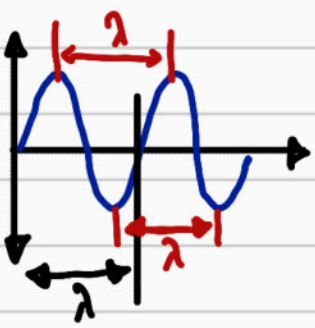


E → Energy of single photon [Joule] or [electron Volt]

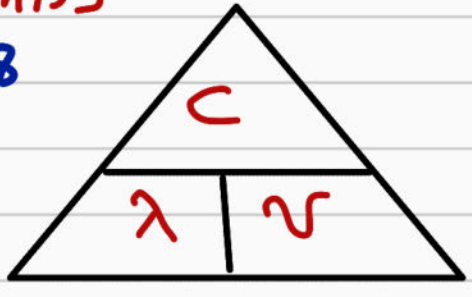
υ_{new} → frequency [Hz] or [1/sec = sec^{-1}]

λ_{lamda} → Wavelength of photon Radiation [m]

C → speed of light in air [3×10^8 m/s]



$C = \lambda \upsilon$ $\quad \lambda = \frac{C}{\upsilon} \quad (C \to 3\times10^8,\ \upsilon \uparrow)$ $\quad \upsilon = \frac{C}{\lambda} \quad (C \to 3\times10^8,\ \lambda \downarrow)$

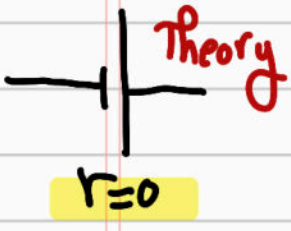


3 types of Bodies:

* Glowing Body: Body which emit heat ① & light ② Sun/Stars/fire/Lamp

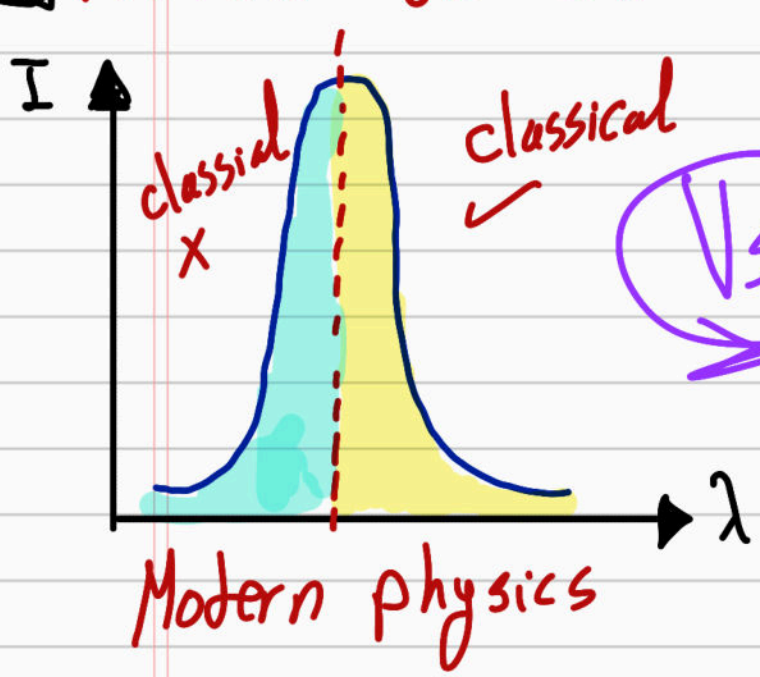
* Non-Glowing Body: Body which absorebs all Radiations falls on it [perfect absorber] but re-emit part of Radiation not all. Earth/Moon/Human

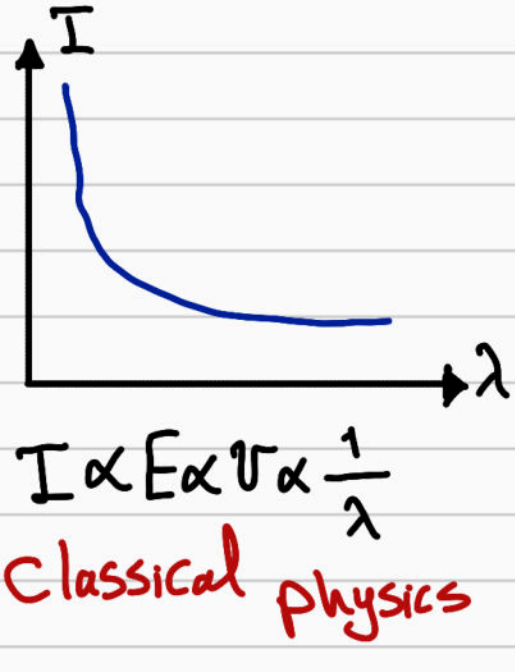
* Black Body: Body which absorebs all Radiations falls on it [perfect absorber] & re-emit all Radiations [perfect re-emitter]. Theory

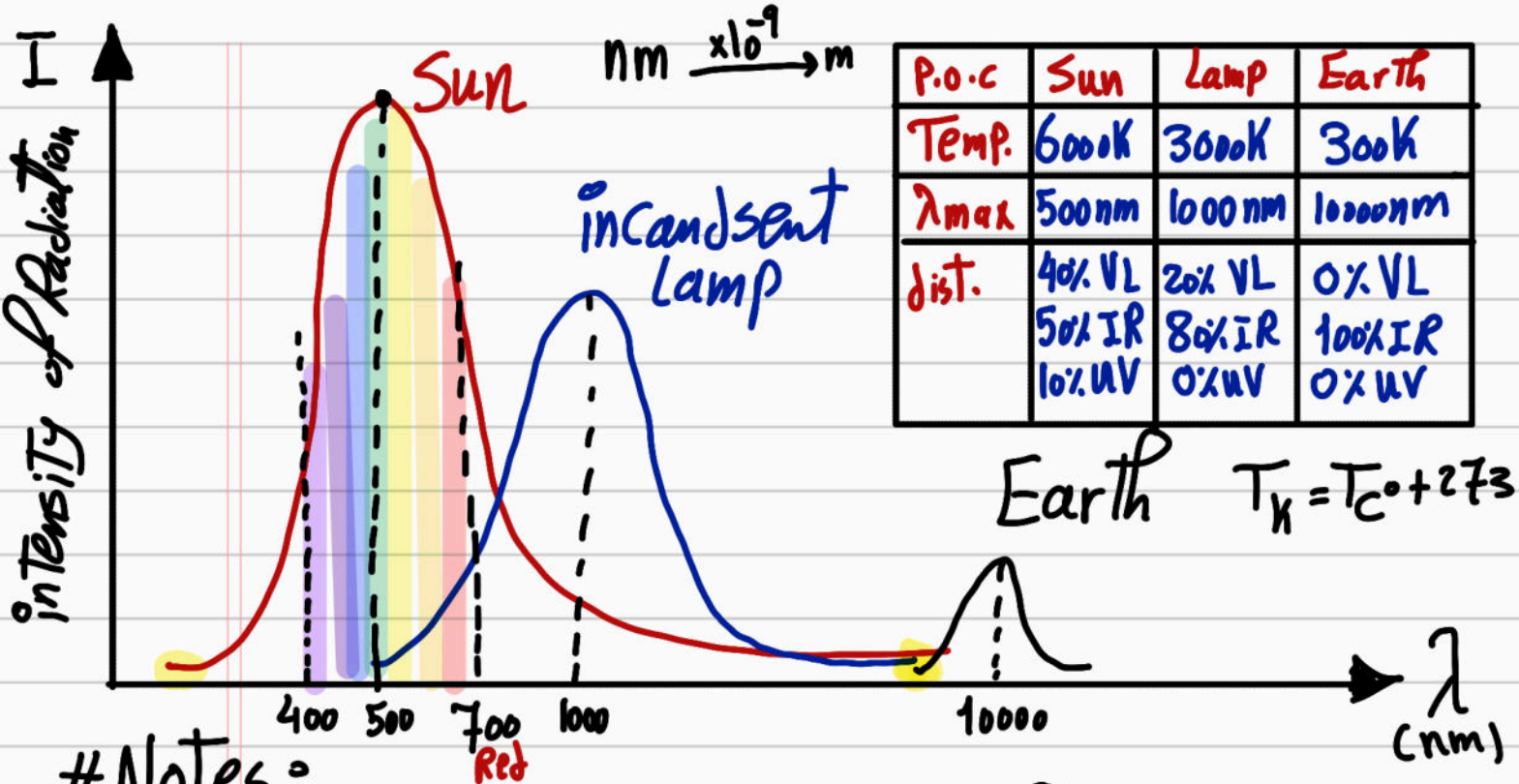




4 Planck's distribution Curve:







P.o.c	Sun	Lamp	Earth
Temp.	6000K	3000K	300K
λ_{max}	500nm	1000nm	10000nm
dist.	40% VL 50% IR 10% UV	20% VL 80% IR 0% UV	0% VL 100% IR 0% UV

$T_K = T_C + 273$

Notes:

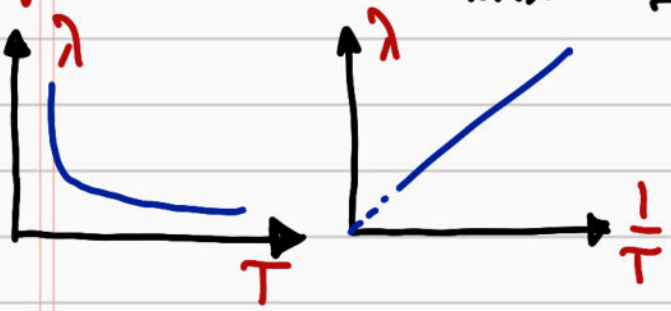
1. Curve akes **bell shape**



2. at high frequincies (low λ) [start of each curve] & low frequincies (high λ) [end of each curve] the intensity of each curte **Tends To Zero.**

3. Wien's LaW $\lambda_{max} \propto \frac{1}{Temp.}$

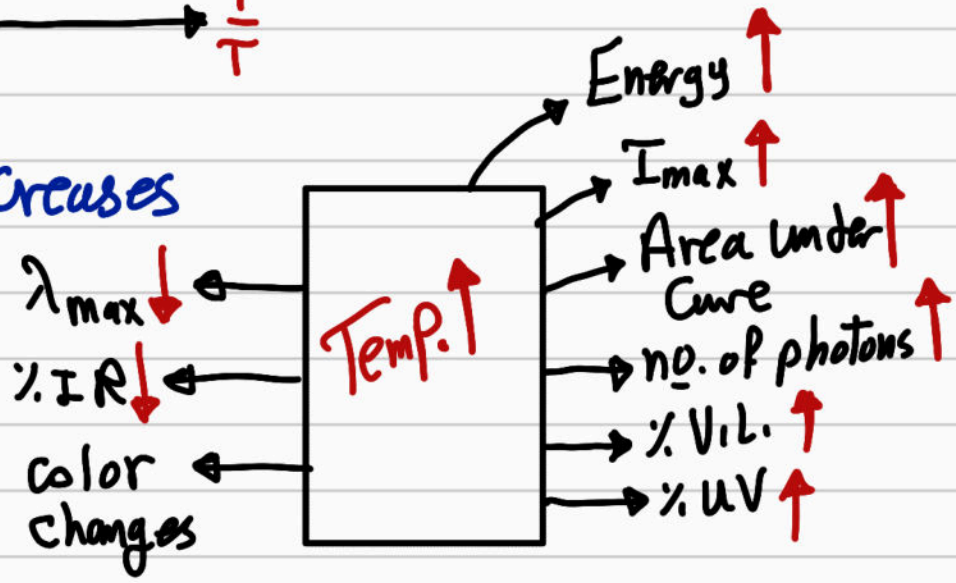
$$\frac{\lambda_1}{\lambda_2} = \frac{T_2}{T_1}$$



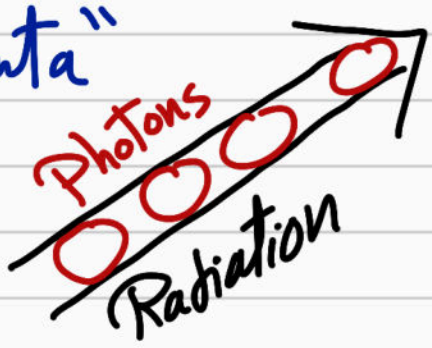
4. as Temp. increases

absoreb more → Temp.↓ dark → Red → orange → yellow → White

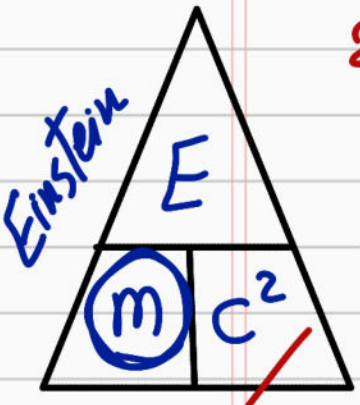
Reflect more → Temp.↑



⑤ Any Radiation consists of small units Called "photons / Quanta"

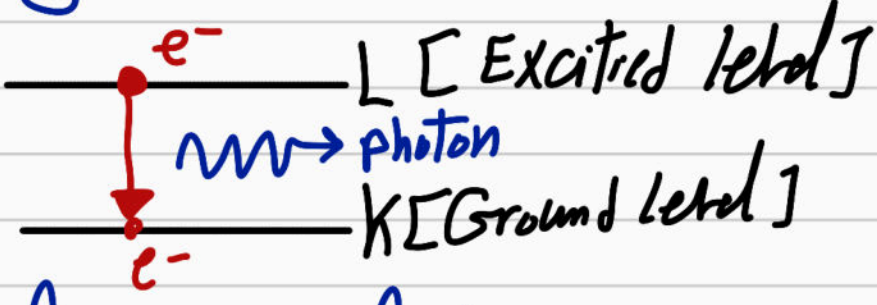


Photons: 1. has no charge*
2. has mass during motion only

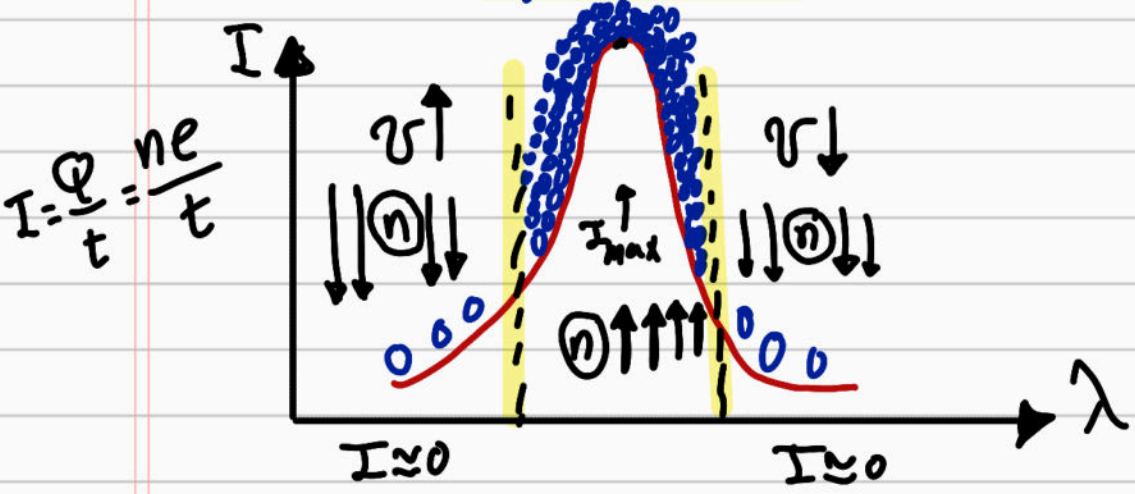


3. When stop its mass converted into energy gained by the object which stops them.
4. Not affected by electric or Magnetic field.

⑥ photons Resulted from Transfering electron from higher energy level To lower energy level.

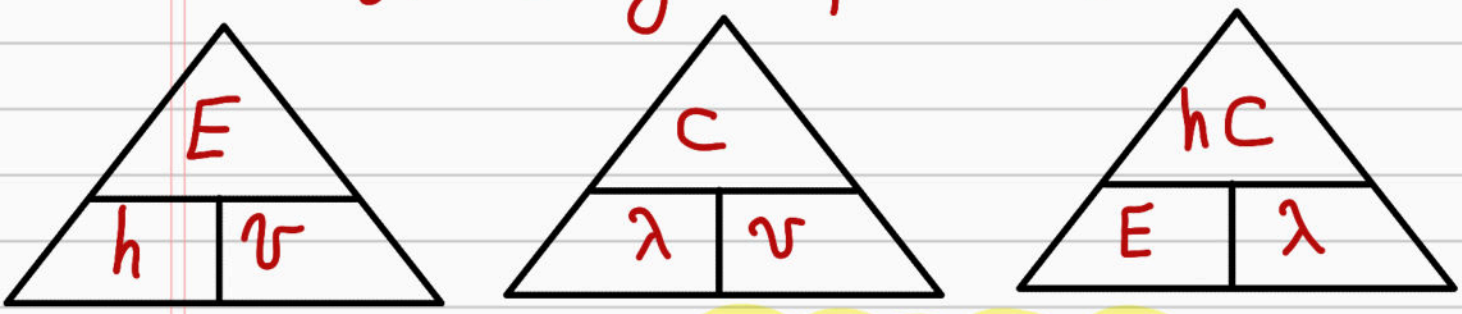


⑦ the energy of beam of photons is Quantized



⑧ Rules Max Planck:

for single photon



h → planck's constant $h = 6.625 \times 10^{-34}\ J.s$

for beam of photons

$$E_{beam} = n \times h\upsilon = n \times \frac{hc}{\lambda}$$

n → no. of photons

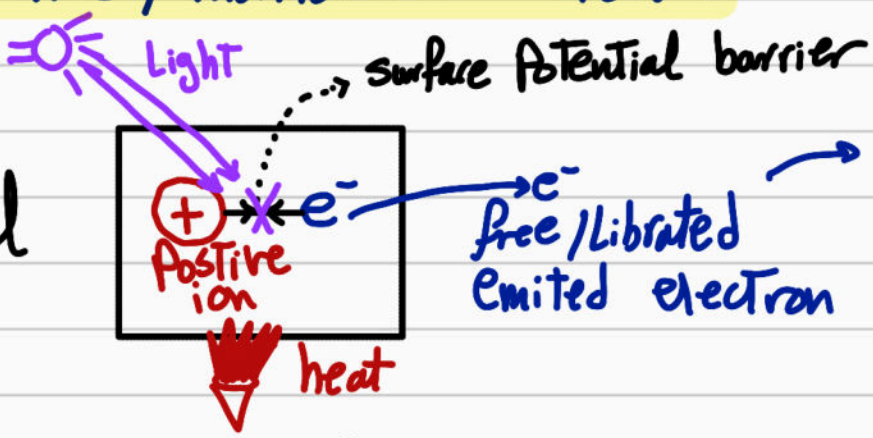
5 Applications:

* Night Vision: IR [Infra Red]
* RADAR: MicroWaves

2 Photoelectric / Thermoionic effect:

*Concept:

any Metal

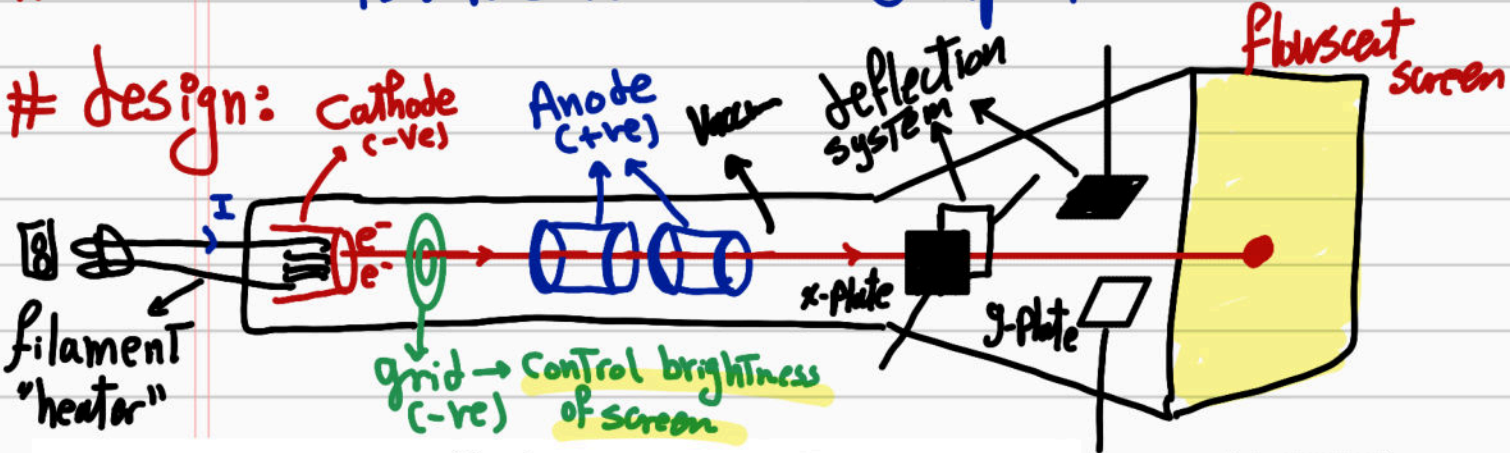


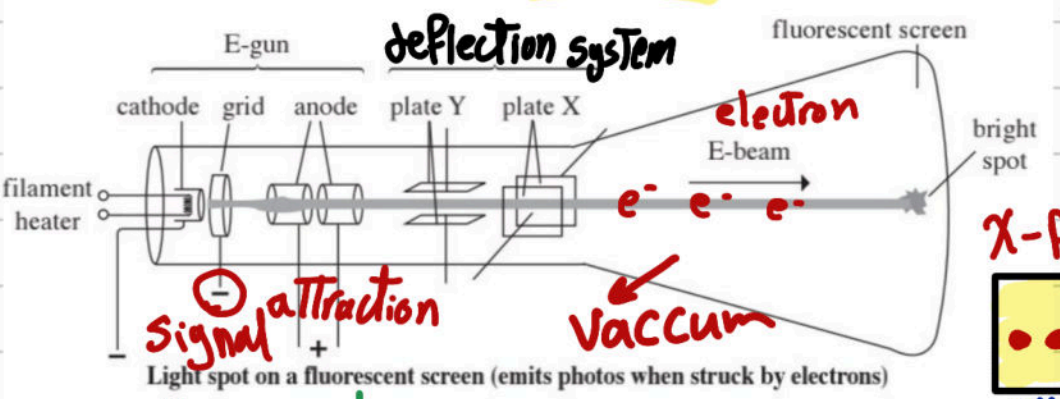
1 Thermo ionic "Cathode Ray Tube [C.R.T.]"

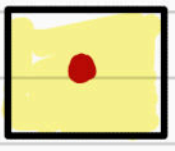
Sci. idea: Thermal ionization

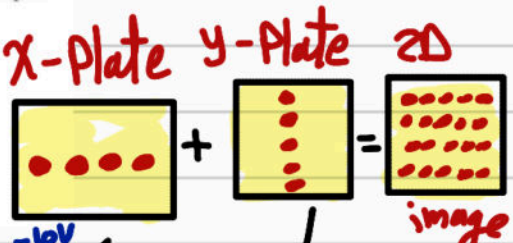
Uses: 1. T.V. Screens 2. Computer monitors

design:









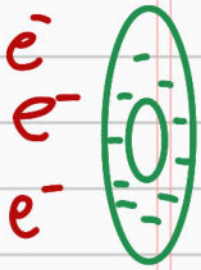
Grid signal −5V ⟶ −10V

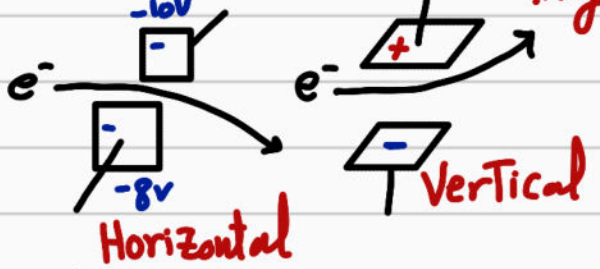
↑ negativity inc.

↑ Repulsion inc.

↓ no. of electrons passing dec.

↓ Brightness / illumnation / Intensity





$KE = \frac{1}{2} m_e v^2 = eV$

$KE \rightarrow$ Kinetic energy [Joule] or [electron Volt]

$m_e \rightarrow$ mass of electron [9.1×10^{-31} Kg]

$v \rightarrow$ Velocity of electrons [m/s]

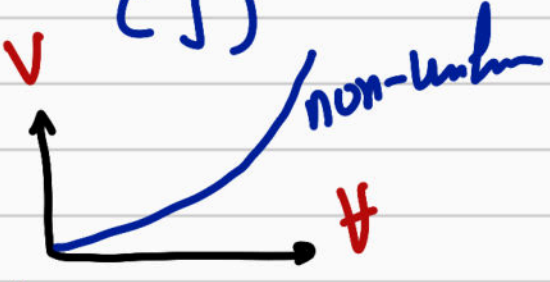
$e^- \rightarrow$ charge of electron [1.6×10^{-19} C]

$V \rightarrow$ Potential difference [Volt] between Cathode & Anode

$$\frac{KE_1}{KE_2} = \left(\frac{v_1}{v_2}\right)^2 = \frac{V_1}{V_2}$$

electron Volt (eV) $\xrightarrow[\times e^-]{\times 1.6 \times 10^{-19} C}$ Joule (J)

$V \propto v^2 \rightarrow \frac{V_1}{V_2} = \left(\frac{v_1}{v_2}\right)^2$



$v \propto \sqrt{V} \rightarrow \frac{v_1}{v_2} = \sqrt{\frac{V_1}{V_2}}$



KE → Joule

$KE = eV$

Joule = Colomb x Volt

$Colomb \times Volt = Kg.m^2.s^{-2}$

Joule

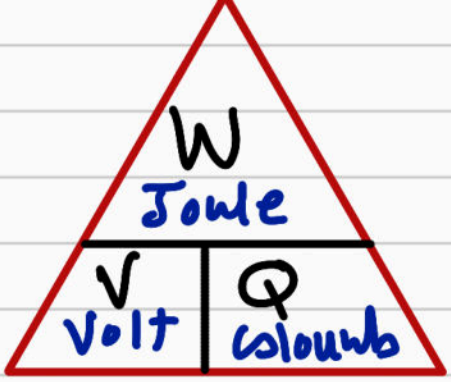
$KE = \frac{1}{2} m_e v^2$

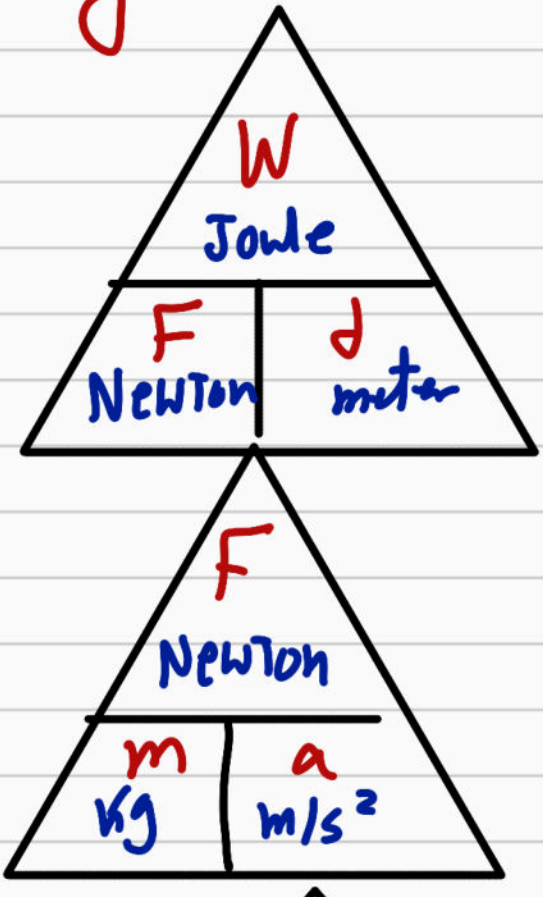
$Joule = Kg.[m/s]^2 = Kg.m^2.s^{-2}$

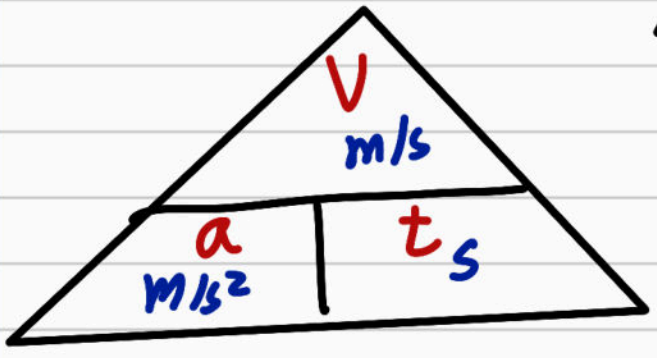
$Joule \equiv N.m$

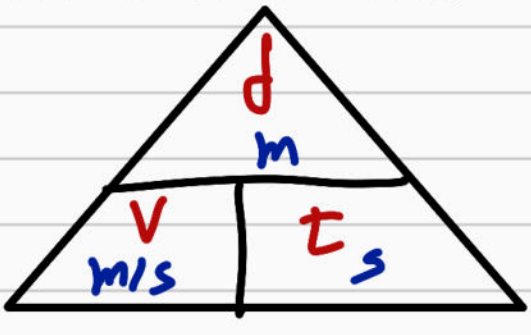
$\equiv (Kg.m/s^2).m$

$\equiv Kg.m^2.s^{-2}$









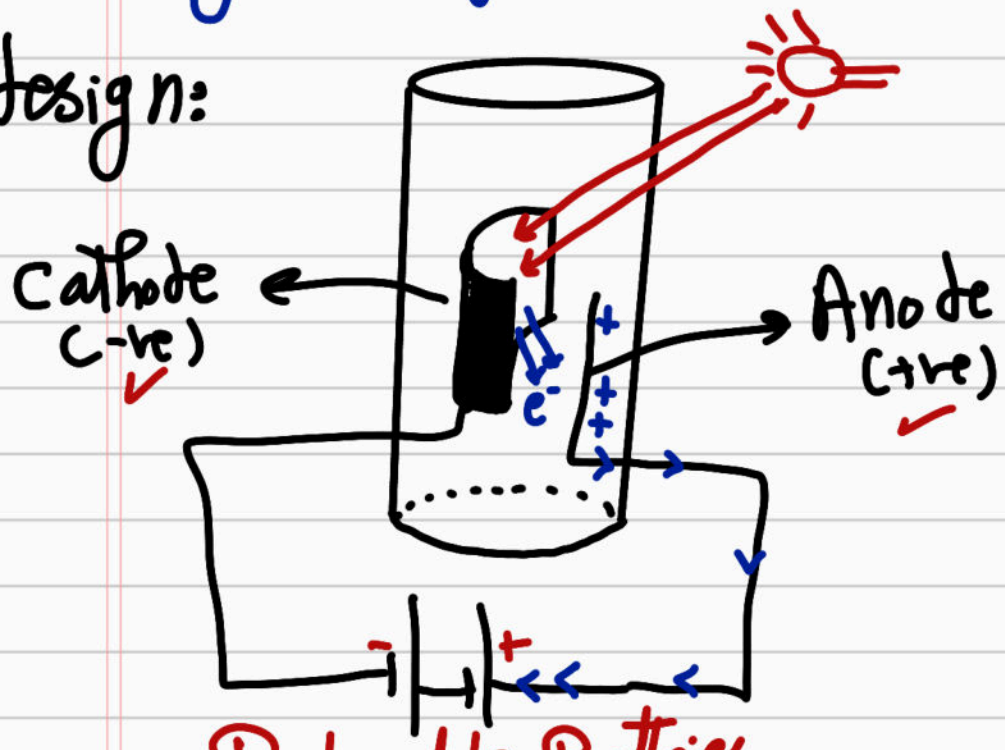
2 Photoelectric phenomena (Sci. Plan)

↳ Photoelectric Cell (كهروضوئية)

uses: Convert

Light energy ⟶ Electric energy

design:



#classical Theory: → Librate/ free

① Any light can emit electrons from any metal surface [Regardless ν_{photon} of incident light]

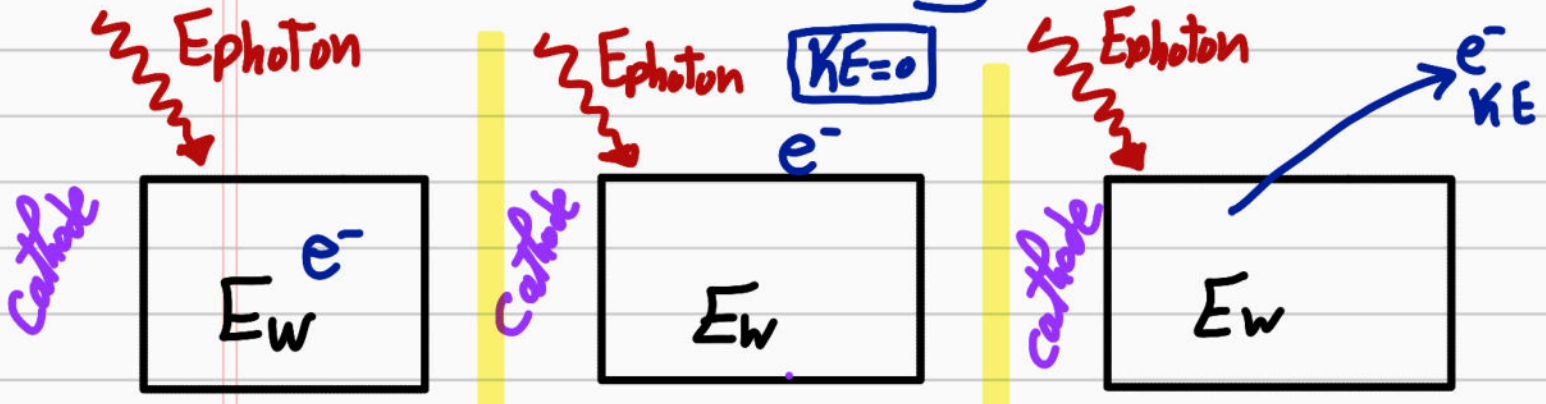
② if can't, increase intensity.

③ if can't, increase exposure Time.

Modern "Einstein" Theory: الفكرة في الطاقة وليس العدد

E_w: Work function: min. amount of energy required To Librate / free / emit electrons from metal surface without having KE.

(Pe: material Cathode)



No emission	Emission Without KE or barely Librated	Emission + KE
$E_{photon} < E_{work}$	$E_{photon} = E_w$	$E_{photon} > E_{work}$
$\upsilon_{photon} < \upsilon_{critical}$	$\upsilon_{photon} = \upsilon_{critical}$	$\upsilon_{photon} > \upsilon_{critical}$
$\lambda_{photon} > \lambda_{critical}$	$\lambda_{photon} = \lambda_{critical}$	$\lambda_{photon} < \lambda_{critical}$

Activiation energy

Cathode material "Pe" =

- E_w → min. amount of energy
- υ_{cr} → min Frequency
- λ_{cr} → Longest Wavelength

required To free / Librate electrons from Cathode material without KE.

incident photon

$E_{photon} = h\nu_{photon} = \frac{hc}{\lambda_{photon}}$

$h = 6.625 \times 10^{-34}$ J.s

$C = 3 \times 10^{8}$ m/s

Barely liberated

$E_W = h\nu_{cr} = \frac{hc}{\lambda_{cr}}$

Cathode material

freed electron e^-

$KE_{e\ max} = \frac{1}{2} m_e \vartheta_{max}^2$

$E_{photon} = E_{work} + KE_{electron}$

*factors:

$E_{ph} \geq E_W$

الحرارة

Temp. ↑

عدد

no. of photons ↑ → no. of electrons ↑ → عدد

or Rate or I

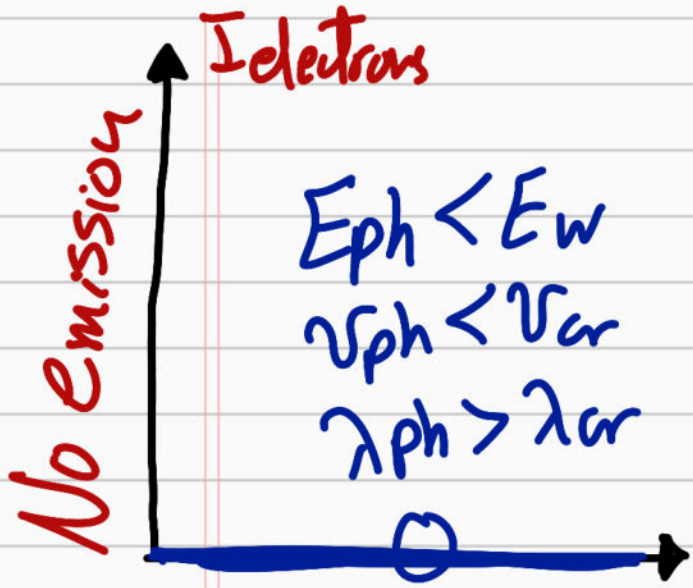
طاقة

$E_{photons}$ ↑ → KE_{e^-} ↑ → زاد

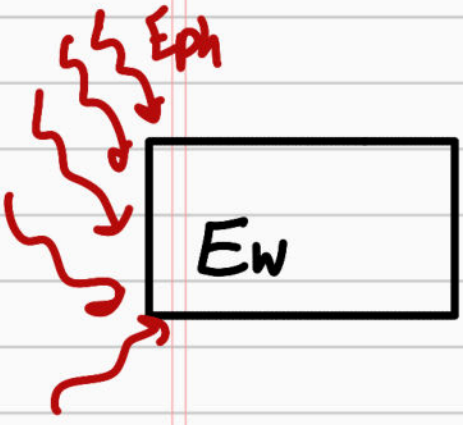
ϑ_{e^-} ↑

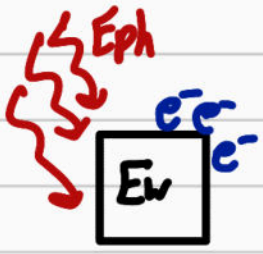
Cathode material

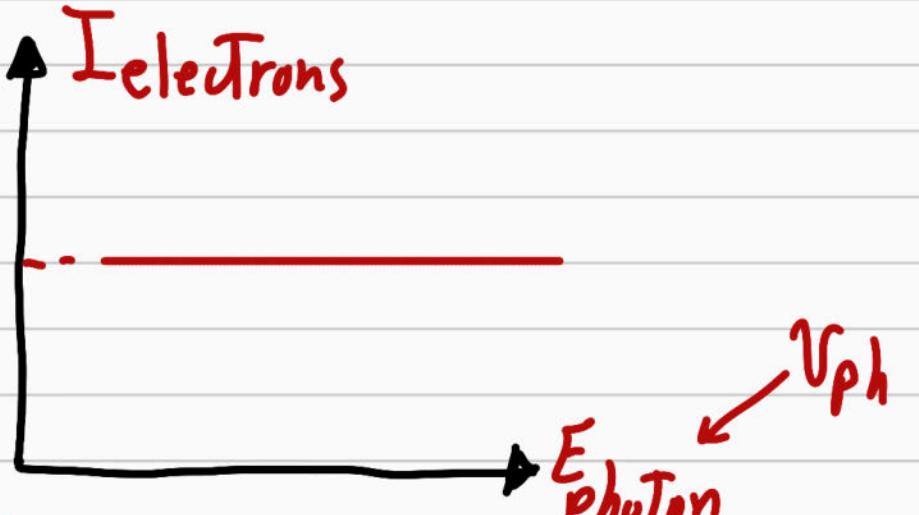
- E_W (Work function)
- ν_{cr} (Critical frequency)
- λ_{cr} (Critical Wavelength)

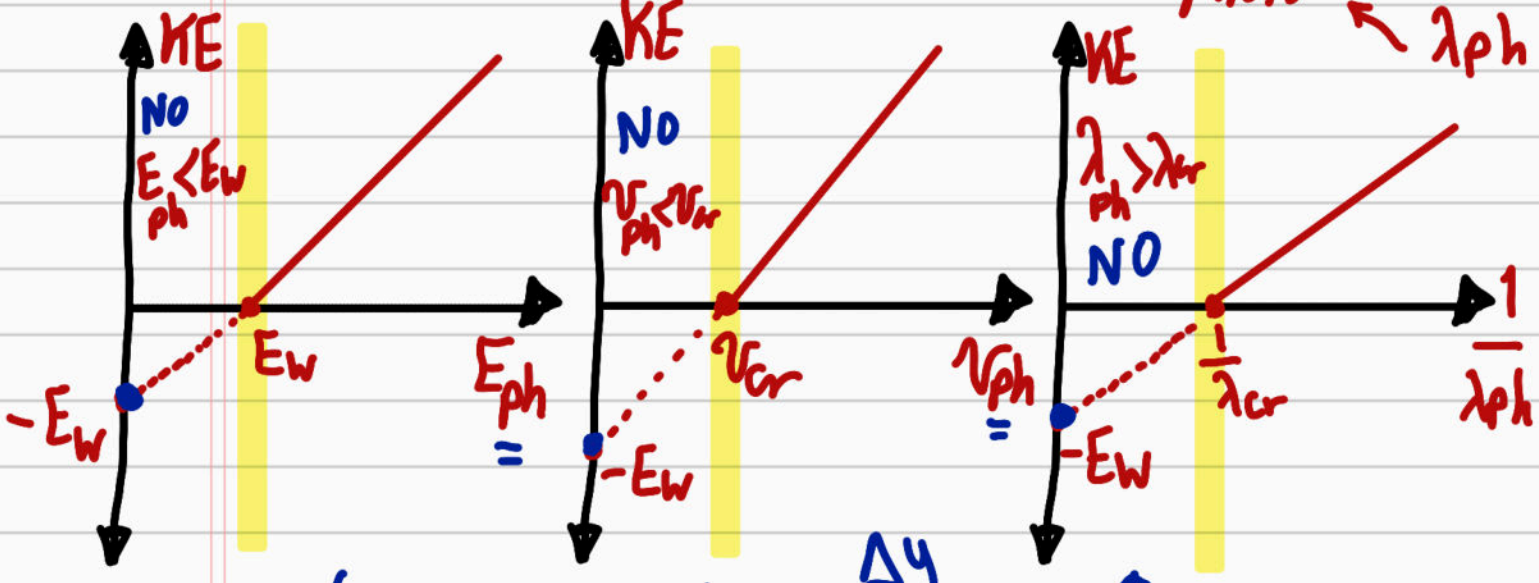












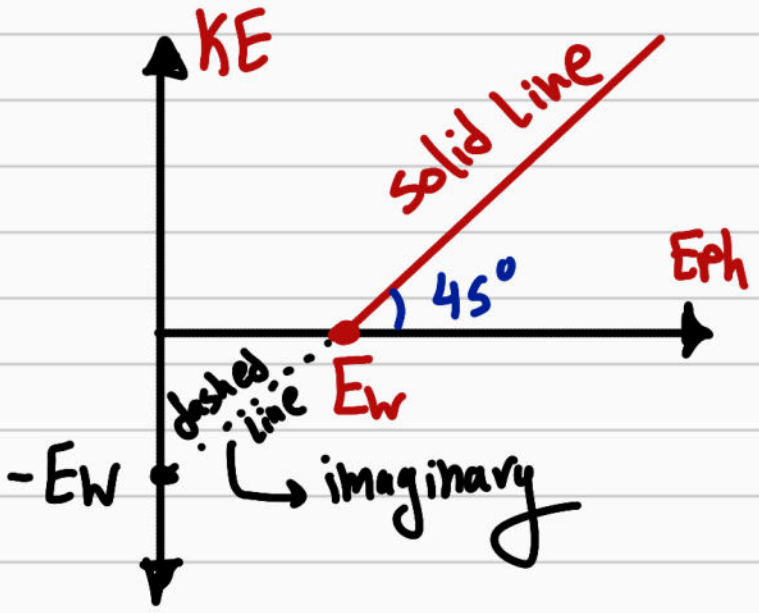
$KE = E_{ph} - E_{work}$ ($E_{work} \to 0$)

$slope = \frac{\Delta y}{\Delta x}$

$E_{ph} = h\nu_{ph}$ (h = const.)

$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{KE}{E_{ph} - E_w}$$

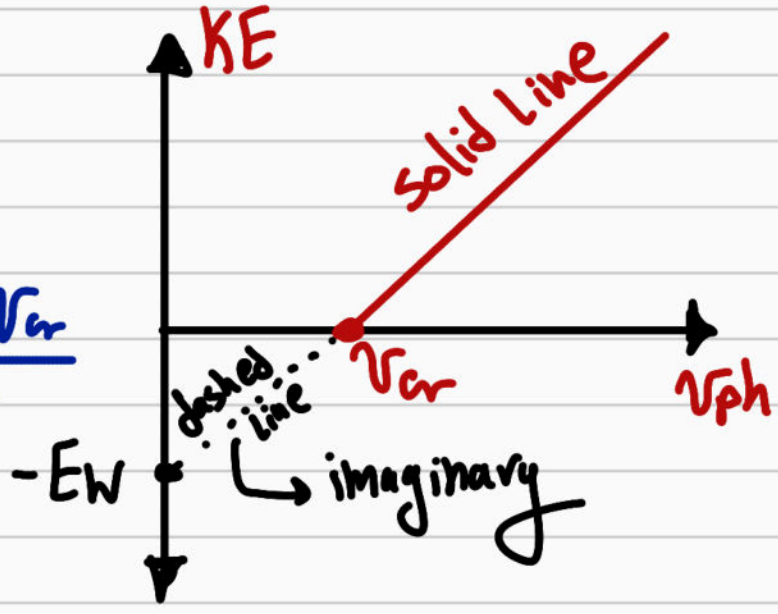
$$= \frac{KE}{KE} = 1$$



$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{KE}{\nu_{ph} - \nu_{cr}}$$

$$= \frac{E_{ph} - E_w}{\nu_{ph} - \nu_{cr}} = \frac{h\nu_{ph} - h\nu_{cr}}{\nu_{ph} - \nu_{cr}}$$

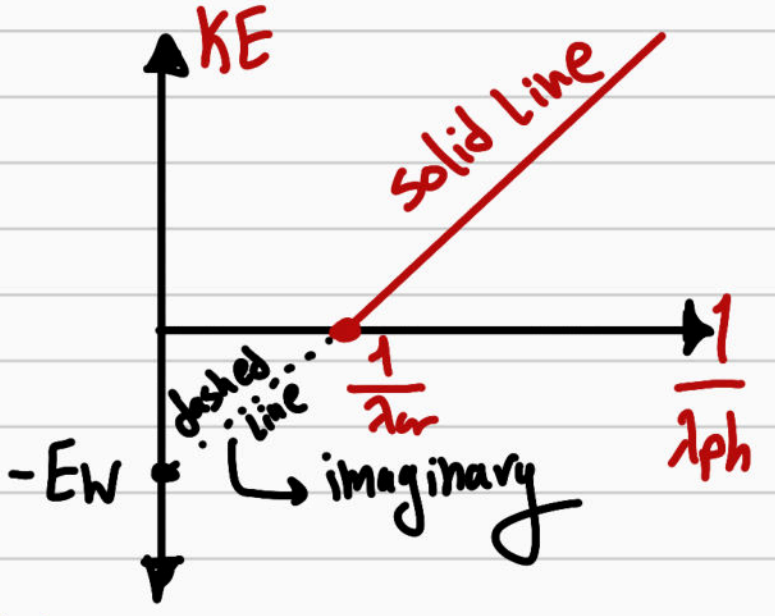
$$= \frac{h[\nu_{ph} - \nu_{cr}]}{[\nu_{ph} - \nu_{cr}]} = h$$



$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{KE}{\frac{1}{\lambda_{ph}} - \frac{1}{\lambda_{cr}}}$$

$$= \frac{E_{ph} - E_w}{\frac{1}{\lambda_{ph}} - \frac{1}{\lambda_{cr}}} = \frac{\frac{hc}{\lambda_{ph}} - \frac{hc}{\lambda_{cr}}}{\frac{1}{\lambda_{ph}} - \frac{1}{\lambda_{cr}}}$$

$$= \frac{hc\left[\frac{1}{\lambda_{ph}} - \frac{1}{\lambda_{cr}}\right]}{\left[\frac{1}{\lambda_{ph}} - \frac{1}{\lambda_{cr}}\right]} = hc$$



Revision

$E_{ph} = h\nu_{ph} = \frac{hc}{\lambda_{ph}}$

$KE_{max} = \frac{1}{2} m_e v^2$

$h = 6.625 \times 10^{-34}$ J.s

↳ planck's const.

$E_w = h\nu_{cr} = \frac{hc}{\lambda_{cr}}$

Cathode material

$E_{ph} = 50J$

e^-

e^-

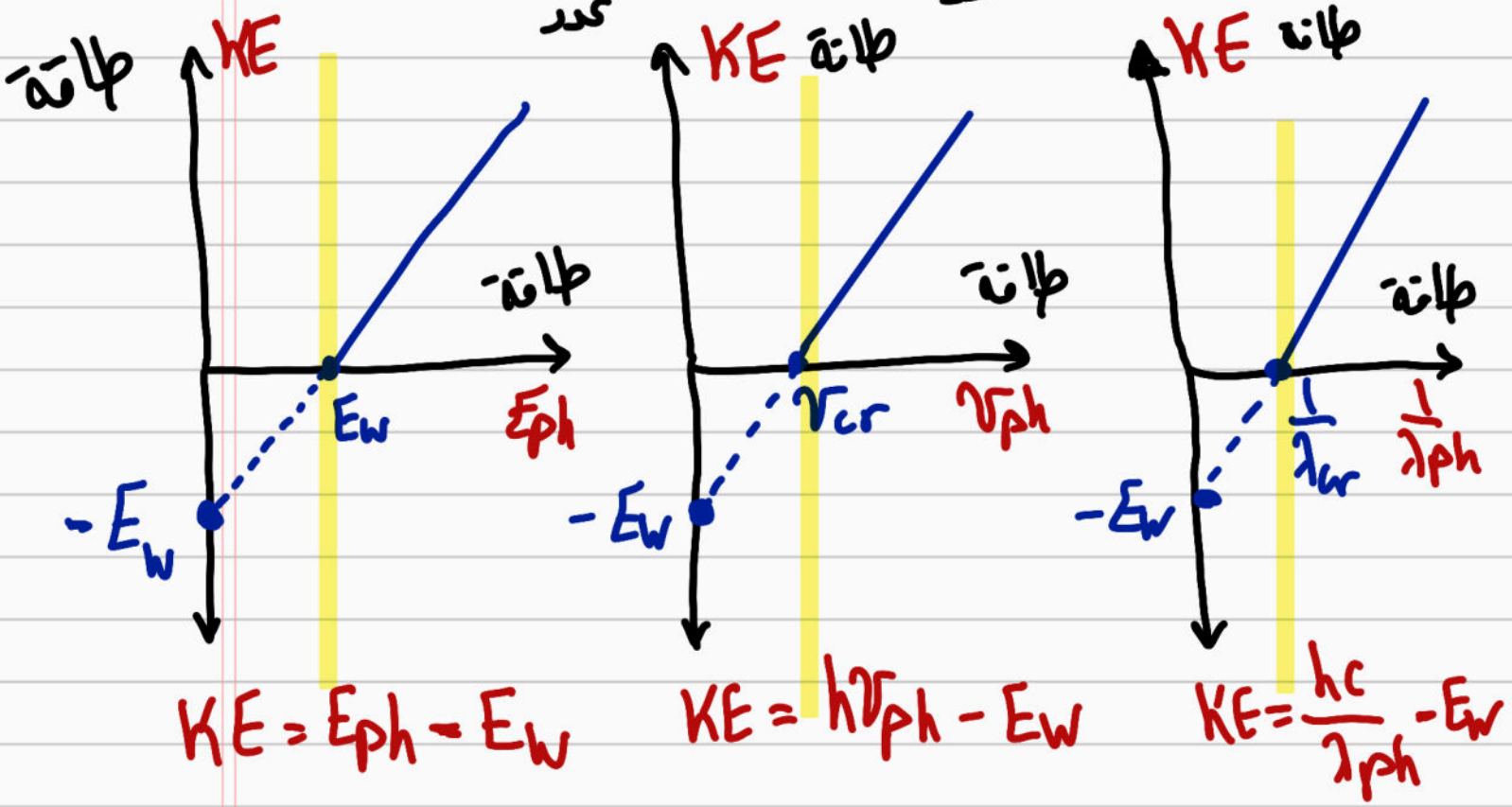
$KE = 40J$

$E_w = 10J$

$E_{ph} = E_w + KE$

الفوتون الضوئي = الشغل + الطاقة الحركية





الطاقة ← تؤثر ← الطاقة

* E_{photon}
* ν_{photon}
* λ_{photon}

* $KE_{electron}$
* $\forall_{electron}$

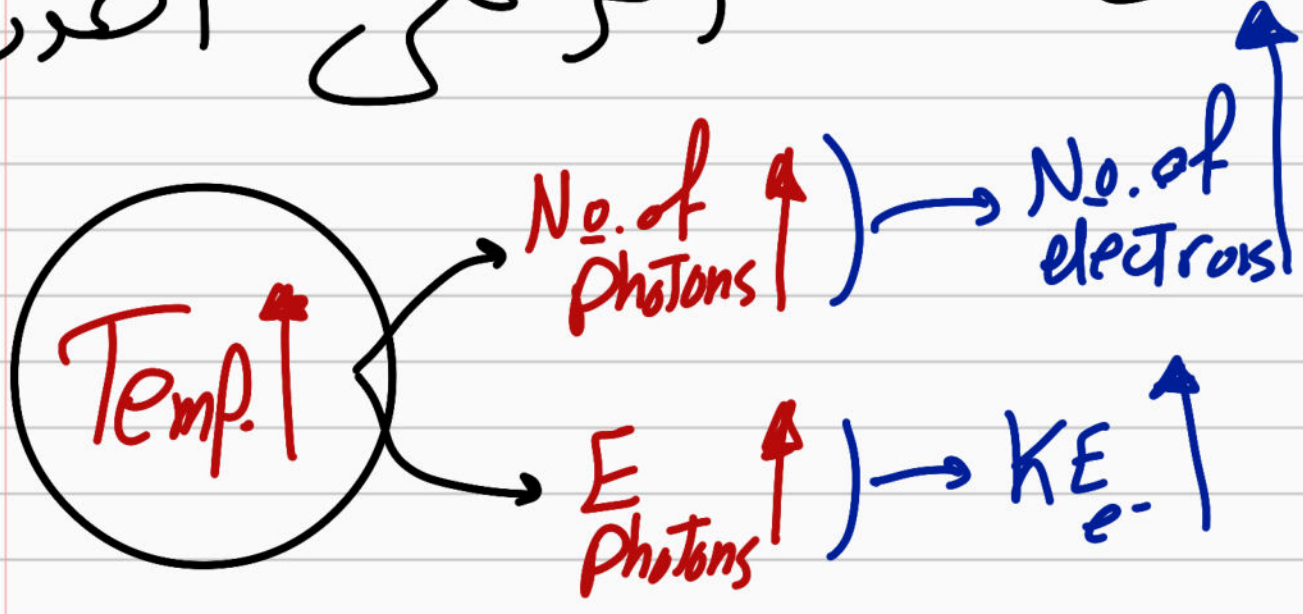
العدد ← يؤثر ← العدد

* I_{photon}
* ϕ_L Rate of incident photon
* $\frac{N_{photons}}{t}$

* $I_{electrons}$
* Rate of electrons emission
* $\frac{N_{e^-}}{t}$

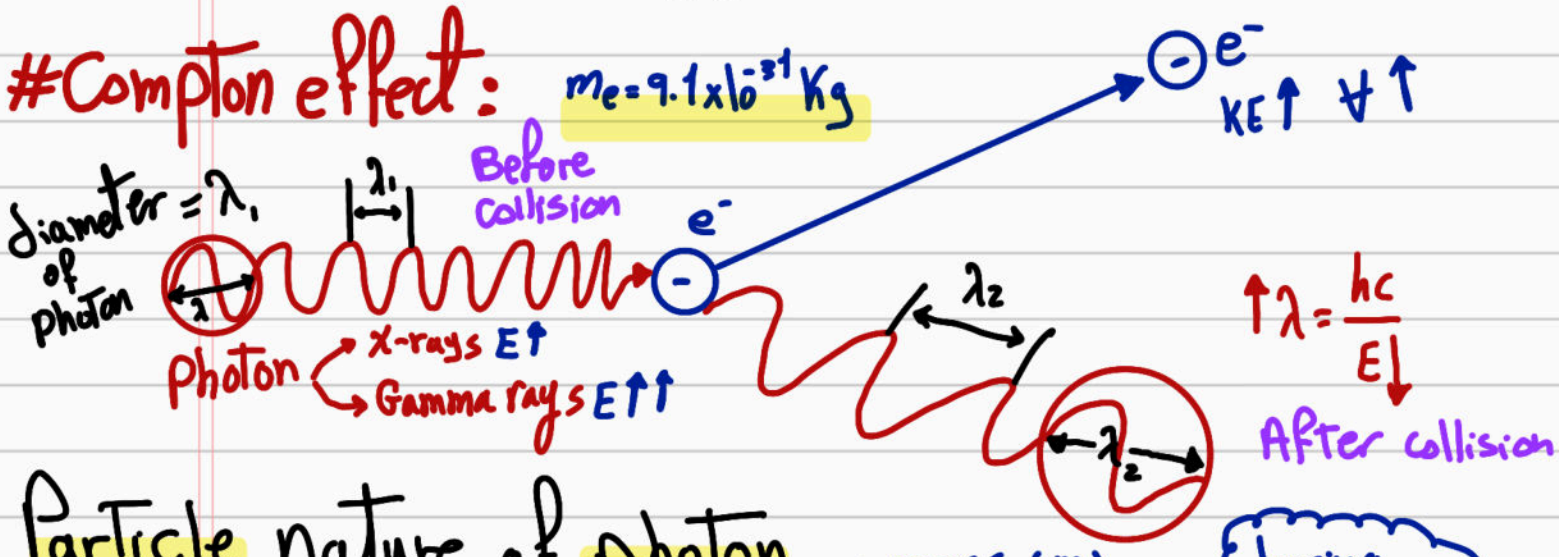
العدد لا يؤثر على الطاقة

الطاقة لا تؤثر على العدد



Lesson ②

#Compton effect: $m_e = 9.1 \times 10^{-31}$ Kg



Particle nature of photon

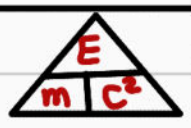
- mass (m)
- momentum (P_L)

during motion

momentum = mass x Speed

$\downarrow P_L = \downarrow m \times c$

#What happen after Collision ??

	Human →	
P.O.C	photon (Wave)	electron (Particles)
mass	$\downarrow m = \frac{E\downarrow}{c^2}$ 	$m_e = 9.1 \times 10^{-31}$ Kg
Speed	$c = 3 \times 10^8$ m/s	Velocity ↑↑↑
momentum	$\downarrow P_L = \downarrow m \times c$	↑↑↑ $P_L = m_e \times v$ ↑↑↑
Wavelength	$\uparrow \lambda = \frac{h}{P_L\downarrow}$	DE-Brogli $\downarrow\downarrow \lambda = \frac{h}{P_L\uparrow\uparrow}$
Volume	↑ Vol ∝ λ ↑	Constant
frequency	$\downarrow \nu = \frac{c \to 3\times10^8}{\lambda\uparrow}$	↑↑ $\nu = \frac{v\uparrow}{\lambda\downarrow}$ -
Energy	$\downarrow E = h\nu\downarrow$	↑↑ $\Delta KE = \frac{1}{2} m_e v^2$ ↑↑
Charge	No charge	$e^- = 1.6 \times 10^{-19}$ C (negative charge)

Law of Conservation of linear momentum:

$$\Sigma P_{L_{before}} = \Sigma P_{L_{after}}$$

$$P_{L_{photon}} + P_{L_{electron}} = P'_{L_{photon}} + P'_{L_{electron}}$$

$$mC + m_e v = m'C + m_e v'$$

decrease — increase

Law of Conservation of Energy:

$$\Sigma E_{before} = \Sigma E_{after}$$

$$E_{photon} + KE_{electron} = E'_{photon} + KE'_{electron}$$

$$mc^2 + \frac{1}{2} m_e v^2 = m'c^2 + \frac{1}{2} m_e (v')^2$$

decreases — increases

Photon Energy before: E_1

Photon Energy after: E_2

$$E_1 + KE_1 = E_2 + KE_2$$

$$E_1 = E_2 + KE_2 - KE_1$$

$$\Delta KE = KE_2 - KE_1$$

$$E_1 = E_2 + \Delta KE$$

$$\Delta KE = E_1 - E_2$$

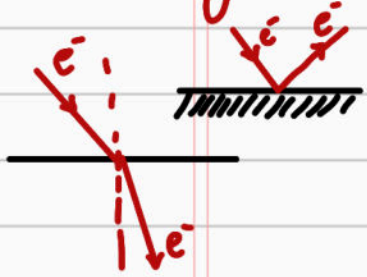
Electron Energy before Collision: KE_1

Electron energy after collision: KE_2

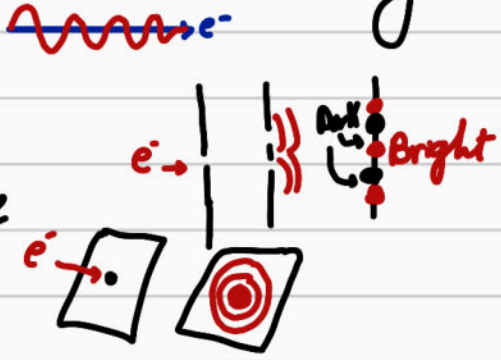
*Compton → Particle nature of photon "during motion"
↳ mass (m) ↳ momentum (P_L)

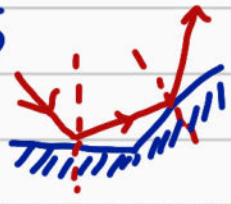
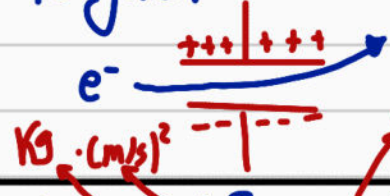
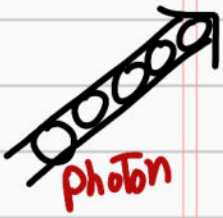
Duality ↳ Wave ↳ Particle

*De-Broglie → Wave nature of electron "during motion"

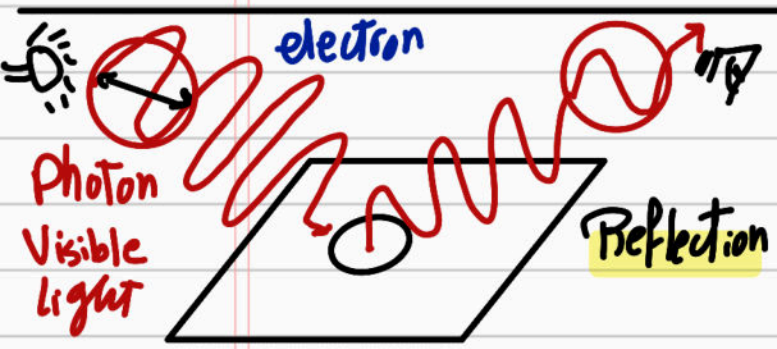


↳ propagation
↳ Reflection
↳ Refraction
↳ Interfance
↳ diffraction



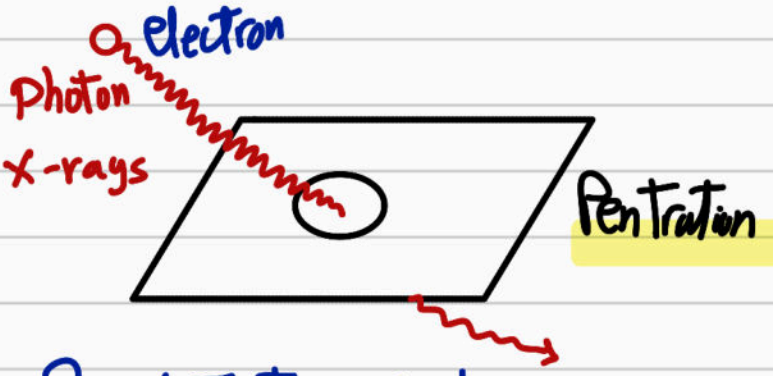
P.O.C	Photon	electron
nature	Wave which has Particle nature	Particle which has Wave nature
charge	has no charge	has negative charge $e^- = 1.6\times10^{-19}C$
its path can be controled using	Glass 	Magnetic or electric field  $\frac{1}{2} m_e v^2 = eV = KE$ (Kg·(m/s)², C, V, J) Can be accelrated $\uparrow v \propto \sqrt{V\uparrow}$
Speed	$c = 3\times10^8$ m/s Cannot be accelrated	
mass 	has mass during motion $m = \frac{E}{c^2}$ When stops $\boxed{m=0}$ its mass is converted into energy gained by the body which stops it.	has Consl nt mass during motion & rest $m_e = 9.1\times10^{-31}$ Kg 

Macroscopic "External"

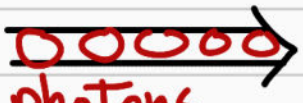
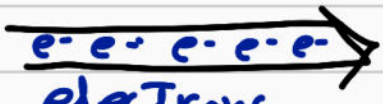


$\lambda >$ Intermolecular spaces distance

Microscopic "Internal"



$\lambda \leqslant$ Intermolecular distance

P.O.C	Light microscope	electron microscope
Used beam	beam of photons  photons	beam of electrons  electrons
Used Lense	Glass	Magnetic or Electric
Magnifying Power	Limited $\lambda_{light} \leqslant$ inter molecular	Unlimited $\downarrow \lambda \propto \frac{1}{v\uparrow}$ $\uparrow v \propto \sqrt{V}\uparrow$
Used Rules	$\lambda = \frac{h}{P_L} = \frac{h}{mc}$	$\downarrow \lambda = \frac{h}{P_L\uparrow} = \frac{h}{m_e v\uparrow}$ de-Broglie $KE = \frac{1}{2} m_e \uparrow v^2 = eV\uparrow$ $

Chapter 5 Rules Summary

1. $\frac{\lambda_1}{\lambda_2} = \frac{T_2(K)}{T_1(K)} \rightarrow \lambda_1 T_1(K) = \lambda_2 T_2(K)$ Wien's law

2. $E_{photon} = h\upsilon_{photon} = \frac{hc}{\lambda_{photon}} = mc^2_{photon} = E_{work} + KE_{electron}$

3. $E_{Work} = h\upsilon_{critical} = \frac{hc}{\lambda_{critical}} = E_{photon} - KE_{electron}$

4. $KE_{electron} = \frac{1}{2} m_e v^2 = eV = E_{photon} - E_{work}$

5. $\lambda_{photon} = \frac{c}{\upsilon_{photon}} = \frac{hc}{E_{photon}} = \frac{h}{P_{L\,photon}}$ de-Broglie

6. $\lambda_{electron} = \frac{h}{P_L} = \frac{h}{m_e v} = \frac{h}{\sqrt{2 m_e KE}} = \frac{h}{\sqrt{2 m_e eV}}$

7. $P_L = \frac{h}{\lambda} = \frac{E_{photon}}{c} = mc_{photon}$ or $m_e v_{electron}$

8. $P_W = \frac{W}{t} \rightarrow \frac{E_{photon}}{t} = E_{photon} \times \boxed{\frac{n}{t}} = E \times \Phi_L$ Rate of incident photons

9. $F = \frac{2 \times P_W}{c}$

10. $\Sigma P_{L\,before} = \Sigma P_{L\,after}$

11. $\Sigma E_{before} = \Sigma E_{after}$